

MALIK HASHIR SHAKEEL

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EDUCATION

NED University of Engineering and Technology Karachi

October 2023 – Present

BE, Computer Systems Engineering (6th Semester) | CGPA: 3.84 / 4.00

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Operating Systems, Computer Architecture, Database Management Systems (DBMS)

SKILLS & INTERESTS

Programming & DSA: Python, C, JavaScript | Data Structures & Algorithms | 270+ LeetCode problems solved across arrays, graphs, dynamic programming, and trees

Machine Learning & Deep Learning: TensorFlow, PyTorch, Scikit-learn | CNNs, RNNs, LSTMs | Model training, fine-tuning, evaluation & optimization

Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn

Web Development: Django, React.js, REST APIs, MySQL, PostgreSQL

Tools & Platforms: Git, GitHub, Linux, Docker

Core Concepts: Operating Systems, System Design, Database Management Systems, Software Engineering Principles

CERTIFICATIONS

• **Deep Learning Specialization** – deeplearning.ai (Coursera) [View Certificate](#)

• **Machine Learning Specialization** – deeplearning.ai (Coursera) [View Certificate](#)

PROJECTS

PII Masking Model

April 2025

GitHub: <https://github.com/MalikHashirShakeel/PII-Masking-Model>

- Designed a production-grade **PII redaction pipeline** targeting **names** and **email addresses** via a **multi-layer hybrid NLP strategy**.
- Fine-tuned **BERT** (bert-base-uncased) on the **WikiNeural** corpus with synthetic email augmentation for the NER task, attaining **F1 ≈ 0.97**.
- Deployed **Qwen2.5-1.5B** as a zero-shot extractor through structured JSON prompting, eliminating the need for task-specific labeled training data.
- Fused regex pattern matching, BERT inference, and LLM fallback into a resilient cascade architecture, maximizing recall on edge cases.
- Established a comprehensive evaluation suite reporting precision, recall, and F1 to support continuous performance monitoring and quality assurance.

Smart Traffic Analytics System

June 2026-Ongoing

GitHub: <https://github.com/MalikHashirShakeel/YOLO-Traffic-Analytics>

- Developed an **end-to-end real-time traffic monitoring system** using **YOLOv8m** for high-accuracy vehicle detection, achieving robust performance on distant and small objects through optimized confidence thresholds and multi-scale inference.
- Implemented **multi-object tracking** with ByteTrack/BoT-SORT, perspective-aware speed estimation, and dynamic density analysis to generate actionable insights including vehicle counts, average speed, traffic flow, and congestion levels.
- Built a comprehensive video analytics pipeline with **real-time visualization**, virtual line counting, adaptive perspective correction, and color-coded congestion alerts (Low → Medium → High → Jam), demonstrating strong system design and practical deployment skills.

Image Captioning Model

March 2026

GitHub: <https://github.com/MalikHashirShakeel/ImageCaptioningModel>

- Engineered an **end-to-end image captioning system** integrating CNN-based visual feature extraction with **LSTM sequence generation** to produce accurate natural-language descriptions.
- Leveraged a **pre-trained convolutional neural network** for high-level image representation, reducing training overhead while preserving feature quality.
- Curated and preprocessed an image-caption dataset to establish robust visual-to-text mappings, improving generalization across diverse inputs.
- Assessed model accuracy using **BLEU** score metrics, with iterative hyperparameter tuning to achieve measurable improvements in caption quality.

ClassConnect — Django Virtual Classroom Management System

March – April 2025

GitHub: <https://github.com/MalikHashirShakeel/ClassConnect>

- Built a scalable, full-stack **virtual classroom platform** using Django, supporting end-to-end academic workflows for educational institutions.
- Delivered core modules encompassing classroom provisioning, assignment lifecycle management, student submission tracking, and threaded discussion boards.
- Enforced **role-based access control** (RBAC) distinguishing instructor and student permissions, ensuring data integrity and secure resource access.